

Arithmetic & Number Sense



Order of operations

- 1 Evaluate $12 + 6 \times (9 - 5) \div 3$.
 - 2 Evaluate $8 + 2 \times 5^2 - 30 \div 6$.
 - 3 Evaluate $\frac{18 - 3 \times 4}{2} + 7$.
 - 4 Evaluate $(7 - 2)^2 - 3 \times (6 - 4)$.
-

Powers and roots

- 5 Evaluate 3^4 .
 - 6 Evaluate $\sqrt{144}$.
 - 7 Evaluate $2^3 \times 2^2$ using the laws of exponents.
 - 8 Evaluate $\sqrt{81} + \sqrt{16}$.
-

Working with fractions

- 9 Compute $\frac{3}{4} + \frac{5}{6}$.
 - 10 Compute $\frac{7}{8} - \frac{1}{3}$.
 - 11 Compute $\frac{2}{3} \times \frac{9}{10}$.
 - 12 Compute $\frac{5}{6} \div \frac{5}{9}$.
 - 13 Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?
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Decimals

- 14 Compute $3.45 + 2.7$.
- 15 Compute $8.6 - 3.25$.

16 Compute 1.2×0.5 .

17 Compute $7.5 \div 0.25$.

Place value and order of magnitude

18 Write 4,500,000 in the form $a \times 10^n$ with $1 \leq a < 10$.

19 Which is greater, 6×10^4 or 58,000?

20 Order from least to greatest: 0.6, $\frac{2}{3}$, 0.65.

Mixed numbers and improper fractions

21 Convert $4\frac{2}{5}$ to an improper fraction.

22 Convert $\frac{47}{6}$ to a mixed number.

23 Compute $2\frac{1}{3} + 1\frac{3}{4}$.

24 Compute $5\frac{1}{2} - 2\frac{2}{3}$.

Absolute value and signed numbers

25 Evaluate $|-7| + |-3|$.

26 Evaluate $-8 + 5 \times (-2)$.

27 Evaluate $|4 - 9| - |2 - 6|$.

Factors, multiples, GCD and LCM

28 Find the greatest common divisor (GCD) of 48 and 60.

29 Find the least common multiple (LCM) of 8 and 12.

30 List all the factors of 36, and state how many there are.

- 31 Is 91 a prime number? Justify your answer.
- 32 Find the greatest common factor of 84 and 126.
-

Divisibility rules

- 33 Without dividing, determine whether 4,578 is divisible by 3.
- 34 Is 2,340 divisible by 9?
- 35 Is 715 divisible by 5?
- 36 Is 1,244 divisible by 4?
-

Estimation and rounding

- 37 Estimate 198×51 by rounding each number to the nearest ten, then find the exact product.
- 38 Round 2.4863 to two decimal places.
- 39 Estimate $\sqrt{50}$ to the nearest whole number.
- 40 A shipment weighs approximately 2,850 kg. Round this to the nearest hundred kilograms.
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Number properties

- 41 If n is an even integer, is $n^2 + n$ always even, always odd, or can it be either?
- 42 Find the remainder when 47 is divided by 6.
- 43 Is 637 divisible by 7? Justify your answer.
- 44 List all prime numbers between 30 and 50.
- 45 If the sum of two integers is odd, is their product always even, always odd, or can it be either?
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Word problems

- 46 A school orders 158 notebooks to be shared as equally as possible among 12 classrooms. How many notebooks does each classroom receive, and how many are left over?
- 47 A caterer must seat 250 guests at tables that each hold 8 people. What is the minimum number of tables required?
- 48 Three friends share a restaurant bill of 186 riyals equally, then each adds a 10-riyal tip. How much does each person pay in total?
- 49 A red light flashes every 8 seconds and a green light flashes every 12 seconds. If both flash together right now, after how many seconds will they next flash together at the same time?
- 50 A water tank holds 480 liters and is filled by a pump running at 15 liters per minute. How long does it take to fill the tank from empty?
- 51 A charity collects 2,340 riyals from 45 equal donations. How much was each donation?
-

MCQ — Order of operations

- 52 Evaluate $20 - 4 \times 3 + 6 \div 2$.
- a. 11
 - b. 27
 - c. 9
 - d. 17
- 53 Evaluate $(5 + 3)^2 - 2 \times 7$.
- a. 78
 - b. 36
 - c. 50
 - d. 46

- 54 Evaluate $\frac{45}{3+6} \times 2$.
- a. 10
 - b. 90
 - c. 7
 - d. 15
- 55 Evaluate $3 + 2 \times (8 - 5)^2$.
- a. 21
 - b. 45
 - c. 17
 - d. 33
- 56 Evaluate $100 - 3 \times (4 + 2 \times 3)$.
- a. 64
 - b. 91
 - c. 88
 - d. 70

MCQ — Exponents and roots

- 57 Evaluate 2^5 .
- a. 16
 - b. 25
 - c. 32
 - d. 10

58 Evaluate $\sqrt{196}$.

a. 98

b. 13

c. 14

d. 15

59 Use the laws of exponents to evaluate $3^2 \times 3^4$.

a. 243

b. 81

c. 729

d. 27

60 Evaluate $\sqrt{225} + \sqrt{64}$.

a. 17

b. 30

c. 23

d. 19

61 Evaluate $5^3 - 4^2$.

a. 109

b. 121

c. 93

d. 45

MCQ — Fraction operations

62 Compute $\frac{2}{5} + \frac{1}{4}$.

a. $\frac{1}{2}$

b. $\frac{11}{20}$

c. $\frac{13}{20}$

d. $\frac{1}{3}$

63 Compute $\frac{5}{6} - \frac{1}{3}$.

a. $\frac{1}{2}$

b. $\frac{1}{3}$

c. $\frac{4}{3}$

d. $\frac{2}{3}$

64 Compute $\frac{3}{4} \times \frac{2}{9}$.

a. $\frac{5}{36}$

b. $\frac{1}{6}$

c. $\frac{2}{3}$

d. $\frac{5}{13}$

65 Compute $\frac{7}{8} \div \frac{7}{4}$.

a. $\frac{1}{2}$

b. $\frac{49}{32}$

c. $\frac{7}{2}$

d. $\frac{1}{4}$

66 Compute $\frac{1}{3} + \frac{1}{4} + \frac{1}{6}$.

a. $\frac{1}{2}$

b. $\frac{2}{3}$

c. $\frac{3}{13}$

d. $\frac{3}{4}$

67 Compute $\frac{5}{9} - \frac{1}{6}$.

a. $\frac{4}{15}$

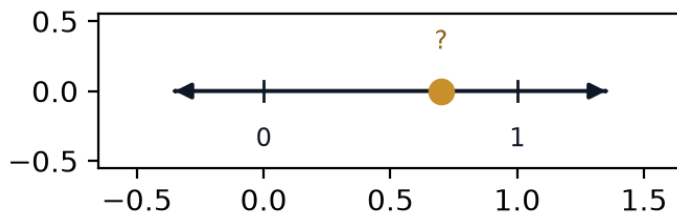
b. $\frac{1}{3}$

c. $\frac{4}{3}$

d. $\frac{7}{18}$

MCQ — Comparing fractions and decimals

68 Which value is largest: the point shown on the number line, $\frac{3}{5}$, or 0.55?



a. The marked point (0.7)

b. $\frac{3}{5}$

c. They are all equal

d. 0.55

69 Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

- a. They are equal
- b. Cannot be determined
- c. $\frac{5}{7}$
- d. $\frac{4}{5}$

70 Order from least to greatest: 0.62 , $\frac{5}{8}$, 0.6 .

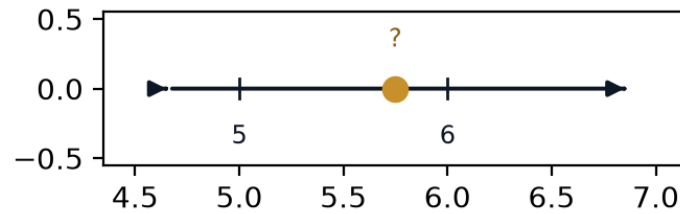
- a. $0.62 < 0.6 < \frac{5}{8}$
- b. $0.6 < \frac{5}{8} < 0.62$
- c. $\frac{5}{8} < 0.6 < 0.62$
- d. $0.6 < 0.62 < \frac{5}{8}$

MCQ — Decimal operations

71 Compute $4.75 + 2.6$.

- a. 7.01
- b. 7.41
- c. 7.35
- d. 6.35

- 72 The number line marks the value of $9.2 - 3.45$. Where does it land?



- a. 5.75
 - b. 6.25
 - c. 5.85
 - d. 5.65
- 73 Compute 1.4×0.6 .
- a. 8.4
 - b. 0.24
 - c. 0.84
 - d. 0.74
- 74 Compute $9.6 \div 0.4$.

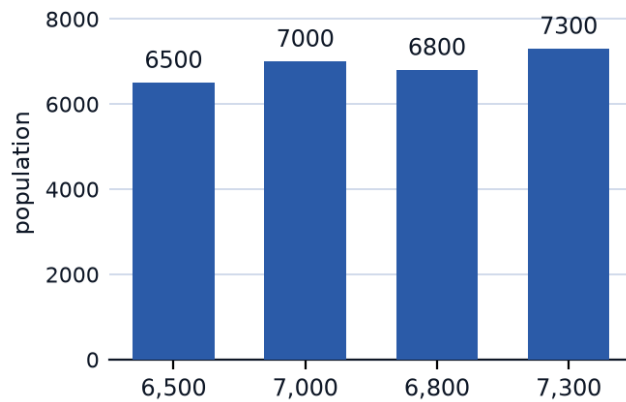
- a. 2.4
- b. 38.4
- c. 24
- d. 0.24

MCQ — Scientific notation & place value

75 Write 6,200,000 in the form $a \times 10^n$ with $1 \leq a < 10$.

- a. 6.2×10^5
- b. 62×10^5
- c. 6.2×10^7
- d. 6.2×10^6

76 The bar chart shows four city populations. Which one equals 7×10^3 ?



- a. 7,000
- b. 6,800
- c. 6,500
- d. 7,300

77 Evaluate 3.4×10^4 as an ordinary number.

- a. 3,400,000
- b. 3,400
- c. 340,000
- d. 34,000

78 In the number 528,437, what is the place value of the digit 8?

- a. Hundred thousands
 - b. Thousands
 - c. Ten thousands
 - d. Hundreds
-

MCQ — Mixed numbers & improper fractions

79 Convert $\frac{29}{6}$ to a mixed number.

- a. $4\frac{5}{6}$
- b. $5\frac{5}{6}$
- c. $5\frac{4}{5}$
- d. $3\frac{5}{6}$

80 Convert $5\frac{3}{7}$ to an improper fraction.

- a. $2\frac{1}{7}$
- b. $3\frac{4}{5}$
- c. $5\frac{3}{7}$
- d. 5

81 Compute $3\frac{1}{4} + 1\frac{5}{6}$.

- a. $5\frac{1}{12}$
- b. 5
- c. $\frac{4}{5}$
- d. $4\frac{11}{12}$

82 Compute $6\frac{1}{5} - 2\frac{3}{4}$.

a. $3\frac{19}{20}$

b. 4

c. $1\frac{17}{20}$

d. $3\frac{9}{20}$

MCQ — Absolute value & signed numbers

83 Evaluate $|-9| + |-4|$.

a. 5

b. -13

c. 13

d. 36

84 Evaluate $-6 + 4 \times (-3)$.

a. -18

b. -30

c. 6

d. -2

85 Evaluate $|5 - 11| - |3 - 8|$.

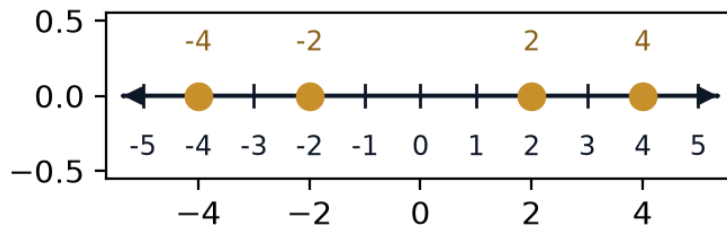
a. 11

b. 3

c. -1

d. 1

- 86 Two of the four points on the number line represent a pair of numbers that are equal in absolute value but opposite in sign. Which pair?



- a. -2 and 2
- b. -4 and 4
- c. -2 and 4
- d. -4 and 2

MCQ — GCD, LCM & factors

- 87 Find the GCD of 54 and 72.

- a. 9
- b. 6
- c. 18
- d. 36

- 88 Find the LCM of 9 and 15.

- a. 3
- b. 24
- c. 45
- d. 135

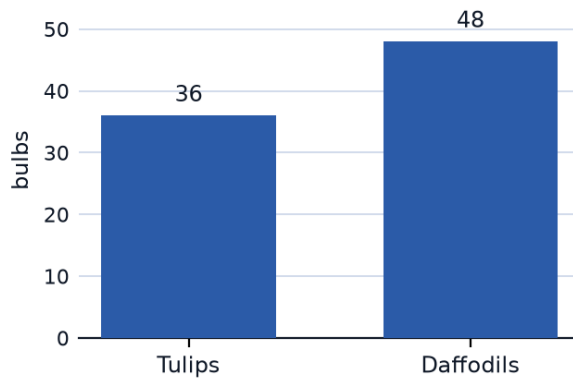
89 How many factors does 60 have?

- a. 6
- b. 8
- c. 12
- d. 10

90 Is 119 a prime number?

- a. No — $119 = 7 \times 17$
- b. Yes, it is prime
- c. No — $119 = 11 \times 9$
- d. Cannot be determined

91 The bar chart shows a gardener's bulb counts. They want to plant them in identical rows with the largest possible number of bulbs per row, each row containing only one type of flower. How many bulbs per row?



- a. 4
- b. 24
- c. 6
- d. 12

MCQ — Divisibility rules

92 Without dividing, is 5,481 divisible by 3?

- a. Cannot tell without dividing
- b. Yes
- c. No
- d. Only by 9, not 3

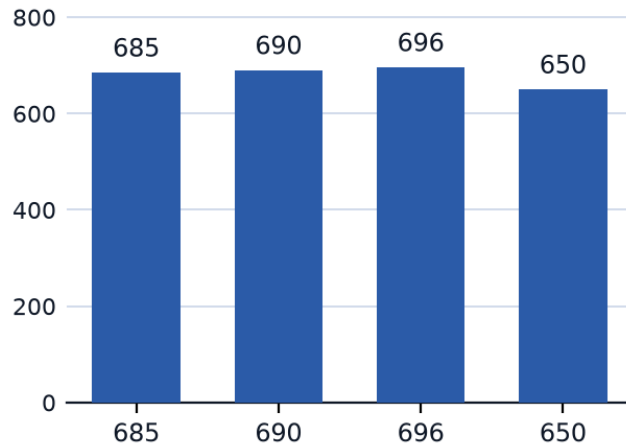
93 Is 2,116 divisible by 4?

- a. Only by 8
- b. Yes
- c. No
- d. Only by 2

94 Is 945 divisible by 9?

- a. Yes
- b. Only by 3
- c. Only by 5
- d. No

- 95 The bar chart shows four candidate numbers. Which one is divisible by both 5 and 6?



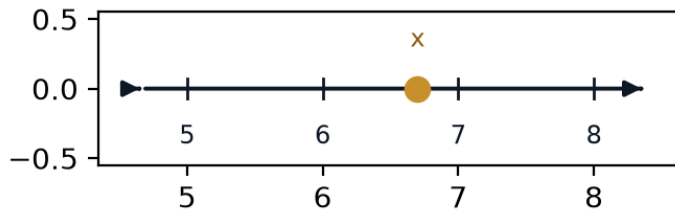
- a. 690
- b. 650
- c. 685
- d. 696

MCQ — Estimation & rounding

- 96 Round 147.362 to the nearest tenth.

- a. 147.3
- b. 150
- c. 147.36
- d. 147.4

- 97 The number line shows the value x rounded to the nearest whole number. If $x = 6.7$, where does the rounded value land?

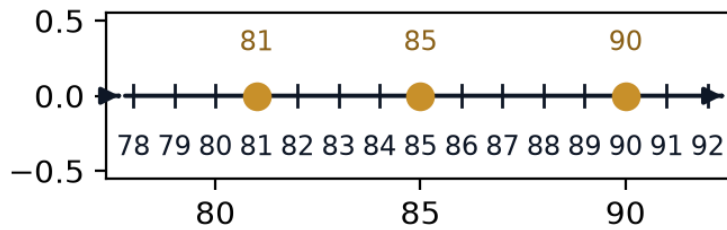


- a. 6
 - b. 6.5
 - c. 8
 - d. 7
- 98 Estimate 312×48 by rounding each factor to the nearest ten, then compare to the exact product 14,976 .
- a. Estimate $310 \times 50 = 15,500$, far from the exact value
 - b. Estimate $300 \times 50 = 15,000$, close to the exact value
 - c. Estimate $320 \times 40 = 12,800$, close to the exact value
 - d. Estimate $310 \times 50 = 15,500$, close to the exact value

MCQ — Number properties

- 99 If n is an odd integer, is $n^2 + n$ always even, always odd, or can it be either?
- a. Can be either
 - b. Depends on the sign of n
 - c. Always odd
 - d. Always even

- 100 The number line marks 85 between consecutive multiples of 9. Find the remainder when 85 is divided by 9.



- a. 3
 - b. 4
 - c. 5
 - d. 76
- 101 List all prime numbers between 50 and 70.
- a. 51, 57, 63, 69
 - b. 53, 59, 61, 67
 - c. 59, 61, 63, 67
 - d. 53, 57, 61, 67

Multiple choice — easy

- 102 Evaluate $(6 + 1)^2 - 8 \times 8$.
- a. -15
 - b. -11
 - c. -19
 - d. -6

103 Evaluate $58 - 9 \times 7 + 12 \div 6$.

- a. 6
- b. -3
- c. -7
- d. 1

104 Evaluate $(8 + 7)^2 - 2 \times 7$.

- a. 211
- b. 220
- c. 207
- d. 215

105 Evaluate $(9 + 2)^2 - 2 \times 2$.

- a. 121
- b. 113
- c. 117
- d. 126

106 Evaluate $55 - 9 \times 4 + 45 \div 5$.

- a. 28
- b. 24
- c. 32
- d. 37

107 Evaluate $(8 + 3)^2 - 4 \times 5$.

- a. 101
- b. 97
- c. 110
- d. 105

108 Evaluate $33 + 8 \times (5 - 1)$.

- a. 65
- b. 69
- c. 61
- d. 74

109 Evaluate $26 + 8 \times (12 - 1)$.

- a. 114
- b. 110
- c. 123
- d. 118

110 Evaluate $34 - 5 \times 6 + 18 \div 6$.

- a. 16
- b. 11
- c. 3
- d. 7

111 Evaluate $15 + 6 \times (8 - 6)$.

a. 27

b. 36

c. 23

d. 31

112 Evaluate $(4 + 4)^2 - 2 \times 2$.

a. 69

b. 64

c. 56

d. 60

113 Evaluate $11 + 3 \times (6 - 4)$.

a. 26

b. 13

c. 17

d. 21

114 Evaluate $59 - 2 \times 5 + 54 \div 6$.

a. 67

b. 54

c. 62

d. 58

115 Evaluate $36 + 5 \times (5 - 1)$.

- a. 56
- b. 60
- c. 65
- d. 52

116 Evaluate $46 - 4 \times 4 + 12 \div 6$.

- a. 32
- b. 36
- c. 28
- d. 41

Multiple choice — medium

117 Compute $\frac{1}{6} \times \frac{6}{7}$.

- a. $\frac{13}{42}$
- b. $\frac{19}{84}$
- c. $-\frac{1}{7}$
- d. $\frac{1}{7}$

118 Compute $\frac{1}{2} - \frac{2}{9}$.

- a. $\frac{1}{6}$
- b. $\frac{5}{18}$
- c. $-\frac{5}{18}$
- d. $\frac{1}{3}$

119 Compute $\frac{4}{9} \div \frac{1}{8}$.

a. $\frac{61}{18}$

b. $-\frac{32}{9}$

c. $\frac{32}{9}$

d. $\frac{131}{36}$

120 Compute $\frac{1}{5} + \frac{1}{5}$.

a. $\frac{3}{5}$

b. $\frac{2}{5}$

c. $-\frac{2}{5}$

d. $\frac{4}{5}$

121 Compute $\frac{6}{8} - \frac{1}{8}$.

a. $\frac{3}{4}$

b. $\frac{3}{8}$

c. $-\frac{5}{8}$

d. $\frac{5}{8}$

122 Compute $\frac{3}{6} - \frac{1}{7}$.

a. $\frac{8}{21}$

b. $-\frac{5}{14}$

c. $\frac{5}{14}$

d. $\frac{13}{42}$

123 Compute $\frac{7}{8} \times \frac{3}{7}$.

a. $-\frac{3}{8}$

b. $\frac{3}{8}$

c. $\frac{11}{24}$

d. $\frac{5}{24}$

124 Compute $\frac{3}{7} \times \frac{4}{6}$.

a. $\frac{31}{84}$

b. $\frac{2}{7}$

c. $\frac{5}{42}$

d. $-\frac{2}{7}$

125 Compute $\frac{1}{3} \div \frac{2}{6}$.

a. $\frac{13}{12}$

b. -1

c. 1

d. $\frac{5}{6}$

126 Compute $\frac{5}{6} - \frac{3}{5}$.

a. $\frac{7}{30}$

b. $\frac{1}{6}$

c. $-\frac{7}{30}$

d. $\frac{4}{15}$

127 Compute $\frac{3}{7} + \frac{3}{4}$.

a. $\frac{31}{28}$

b. $\frac{33}{28}$

c. $-\frac{33}{28}$

d. $\frac{17}{14}$

128 Compute $\frac{1}{4} \div \frac{7}{8}$.

a. $\frac{2}{7}$

b. $-\frac{2}{7}$

c. $\frac{31}{84}$

d. $\frac{5}{42}$

129 Compute $\frac{8}{9} + \frac{7}{9}$.

a. $\frac{13}{9}$

b. $-\frac{5}{3}$

c. $\frac{16}{9}$

d. $\frac{5}{3}$

130 Compute $\frac{3}{8} + \frac{1}{9}$.

a. $\frac{1}{2}$

b. $\frac{35}{72}$

c. $-\frac{35}{72}$

d. $\frac{11}{24}$

131 Compute $\frac{3}{7} \div \frac{7}{9}$.

- a. $\frac{113}{294}$
- b. $\frac{373}{588}$
- c. $-\frac{27}{49}$
- d. $\frac{27}{49}$

Multiple choice — hard

132 Find the greatest common divisor (GCD) of 90 and 72.

- a. 36
- b. 18
- c. 21
- d. 15

133 Find the greatest common divisor (GCD) of 143 and 117.

- a. 16
- b. 13
- c. 26
- d. 10

134 Find the greatest common divisor (GCD) of 132 and 88.

- a. 44
- b. 88
- c. 41
- d. 47

135 Find the greatest common divisor (GCD) of 99 and 55.

- a. 14
- b. 22
- c. 8
- d. 11

136 Find the greatest common divisor (GCD) of 36 and 81.

- a. 9
- b. 6
- c. 12
- d. 18

137 Find the least common multiple (LCM) of 19 and 16.

- a. 310
- b. 304
- c. 305
- d. 303

138 Find the least common multiple (LCM) of 15 and 12.

- a. 63
- b. 57
- c. 60
- d. 180

139 Find the least common multiple (LCM) of 4 and 8.

- a. 4
- b. 12
- c. 8
- d. 32

140 Find the least common multiple (LCM) of 9 and 13.

- a. 123
- b. 116
- c. 117
- d. 118

141 Find the least common multiple (LCM) of 11 and 14.

- a. 160
- b. 155
- c. 154
- d. 153

142 A shipment of 277 pencils is divided as equally as possible among 15 boxes. How many pencils does each box receive, and how many are left over?

- a. 18, 7
- b. 18, 8
- c. 7, 18
- d. 19, 7

- 143** A shipment of 310 books is divided as equally as possible among 15 shelves. How many books does each shelf receive, and how many are left over?
- a. 20, 11
 - b. 21, 10
 - c. 10, 20
 - d. 20, 10
- 144** A shipment of 147 stickers is divided as equally as possible among 15 students. How many stickers does each student receive, and how many are left over?
- a. 9, 12
 - b. 9, 13
 - c. 10, 12
 - d. 12, 9
- 145** A shipment of 336 chairs is divided as equally as possible among 23 rows. How many chairs does each row receive, and how many are left over?
- a. 16, 13
 - b. 14, 14
 - c. 15, 14
 - d. 14, 15
- 146** A shipment of 253 cups is divided as equally as possible among 14 bags. How many cups does each bag receive, and how many are left over?
- a. 1, 18
 - b. 19, 1
 - c. 18, 2
 - d. 18, 1